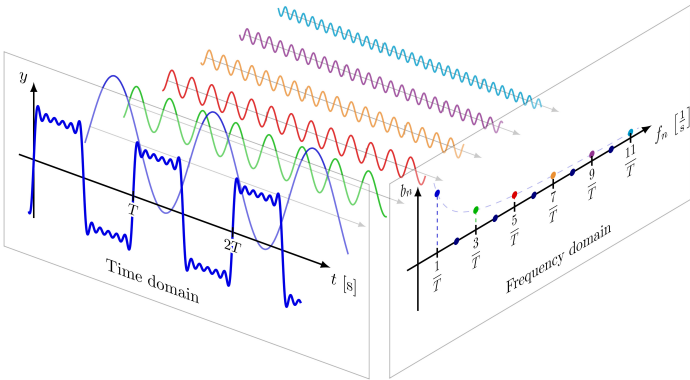








$$P_{x^{\nu}}\nu$$



$$\hat{f}(\omega)f(t)$$

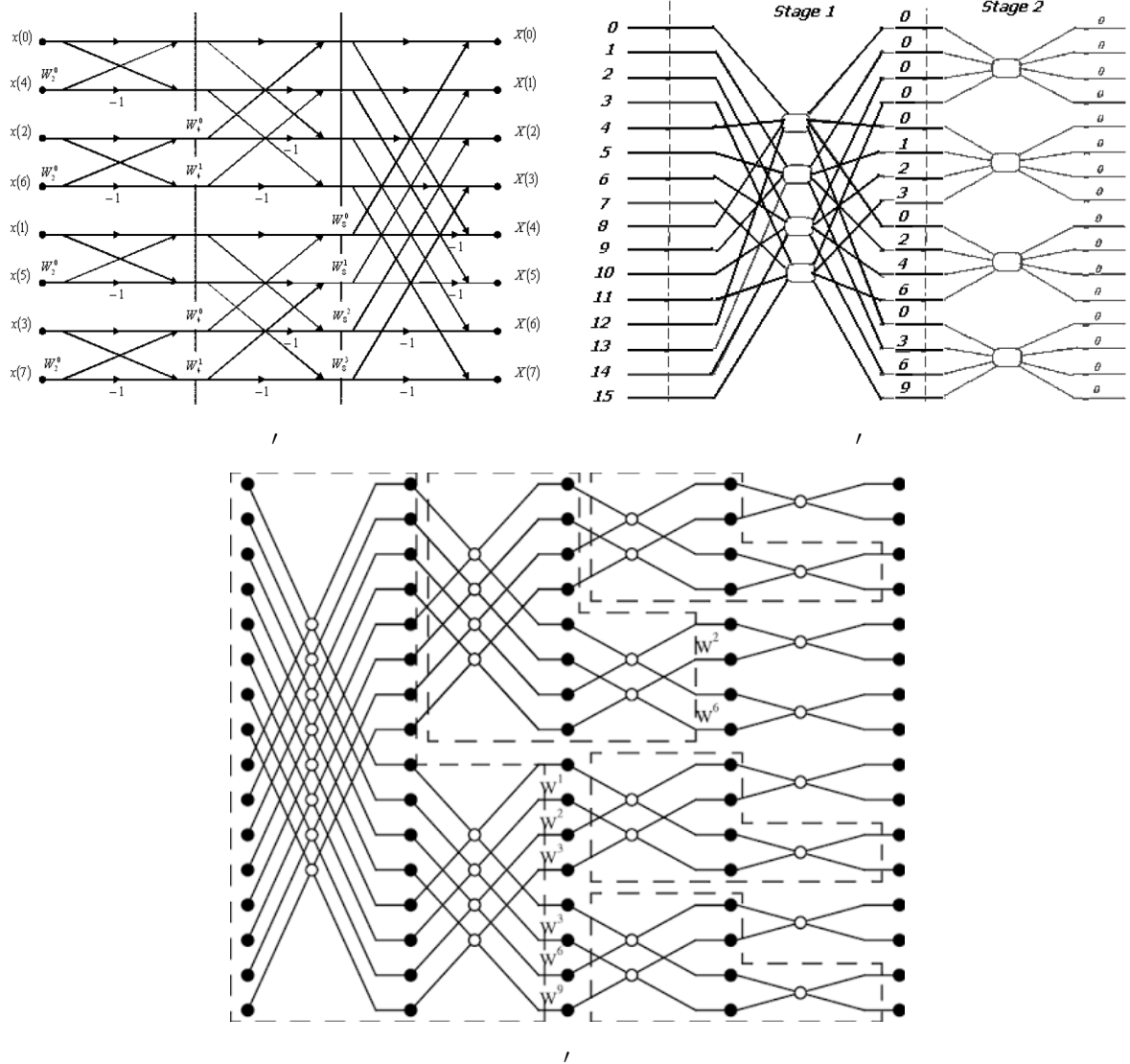
$$\hat{f}(\omega)=\int_{-\infty}^{\infty}f(t)e^{-i\omega t}\,dt$$

$$O(N^2)O(N\log(N))$$

$$N$$

$$N$$

$$N$$



$$x[n]N$$

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot W_N^{kn}, W_N = e^{-j\frac{2\pi}{N}}$$

$$W_N$$

$$x[n]e[n] = x[2n]o[n] = x[2n + 1]$$

$$X[k] = \sum_{n=0}^{N/2-1} x[2n] \cdot W_N^{k \cdot 2n} + \sum_{n=0}^{N/2-1} x[2n + 1] \cdot W_N^{k \cdot (2n+1)}$$

$$W_N^{2kn} = W_{N/2}^{kn}$$

$$X[k] = \underbrace{\sum_{n=0}^{N/2-1} e[n] \cdot W_{N/2}^{kn}}_{(E[k])} + W_N^k \cdot \underbrace{\sum_{n=0}^{N/2-1} o[n] \cdot W_{N/2}^{kn}}_{(O[k])}$$

$$E[k]O[k]$$

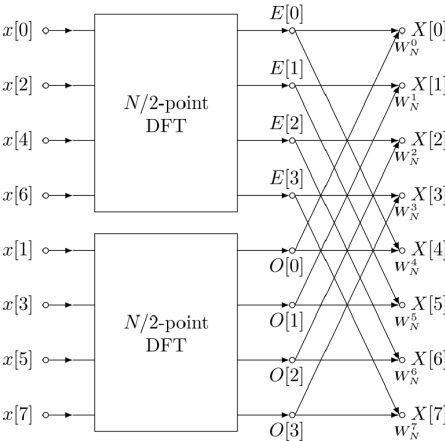
$$X[k] = E[k] + W_N^k \cdot O[k]$$

$$Nk = 0,1,\ldots,N-1$$

$$X[k] = E[k] + W_N^k \cdot O[k]$$

$$X\left[k+\frac{N}{2}\right] = E[k] - W_N^k \cdot O[k]$$

$$k=0,1,\ldots,N/2-1\quad N=2^mm=\log_2(N)$$



$$NN=4^m$$

$$x[n]$$

$$x_0[n] = x[4n], \qquad \qquad \qquad x_1[n] = x[4n+1],$$

$$x_2[n] = x[4n+2], \qquad \qquad \qquad x_3[n] = x[4n+3]$$

$$X[k] = \sum_{n=0}^{N/4-1} \left[x[4n] + x[4n+1]W_N^k + x[4n+2]W_N^{2k} + x[4n+3]W_N^{3k} \right] W_{N/4}^{kn}$$

$$N/4X_0, X_1, X_2, X_3$$

$$X[k] = X_0[k] + W_N^k X_1[k] + W_N^{2k} X_2[k] + W_N^{3k} X_3[k]$$

$$A = X_0[k], B = W_N^k X_1[k], C = W_N^{2k} X_2[k], D = W_N^{3k} X_3[k]$$

$$\begin{aligned} X[k] &= A + B + C + D \\ X\left[k + \frac{N}{4}\right] &= A - jB - C + jD \\ X\left[k + \frac{N}{2}\right] &= A - B + C - D \\ X\left[k + \frac{3N}{4}\right] &= A + jB - C - jD \end{aligned}$$

$$\log_4(N)$$

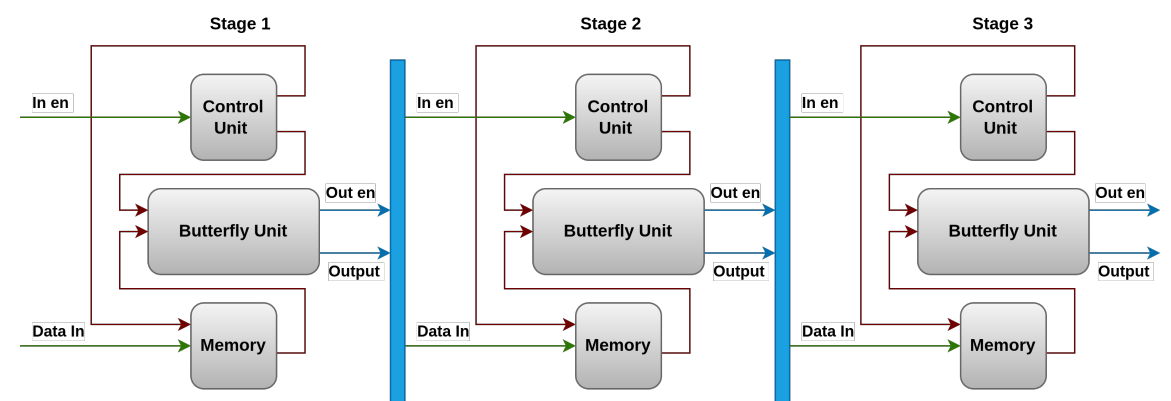
$$X[k] = \sum_{n=0}^{N/2-1} x[2n]W_{N/2}^{kn} + \sum_{n=0}^{N/4-1} x[4n+1]W_N^{k(4n+1)} + \sum_{n=0}^{N/4-1} x[4n+3]W_N^{k(4n+3)}$$

$$X[k] = \sum_{n=0}^{N/2-1} x[2n]W_{N/2}^{kn} + W_N^k \sum_{n=0}^{N/4-1} x[4n+1]W_{N/4}^{kn} + W_N^{3k} \sum_{n=0}^{N/4-1} x[4n+3]W_{N/4}^{kn}$$

$$E[k]N/2O_1[k]O_3[k]N/4$$

$$\begin{aligned} X[k] &= E[k] + (W_N^k O_1[k] + W_N^{3k} O_3[k]) \\ X\left[k + \frac{N}{2}\right] &= E[k] - (W_N^k O_1[k] + W_N^{3k} O_3[k]) \\ X\left[k + \frac{N}{4}\right] &= E\left[k + \frac{N}{4}\right] - j(W_N^k O_1[k] - W_N^{3k} O_3[k]) \\ X\left[k + \frac{3N}{4}\right] &= E\left[k + \frac{N}{4}\right] + j(W_N^k O_1[k] - W_N^{3k} O_3[k]) \end{aligned}$$

$$N=4^m$$

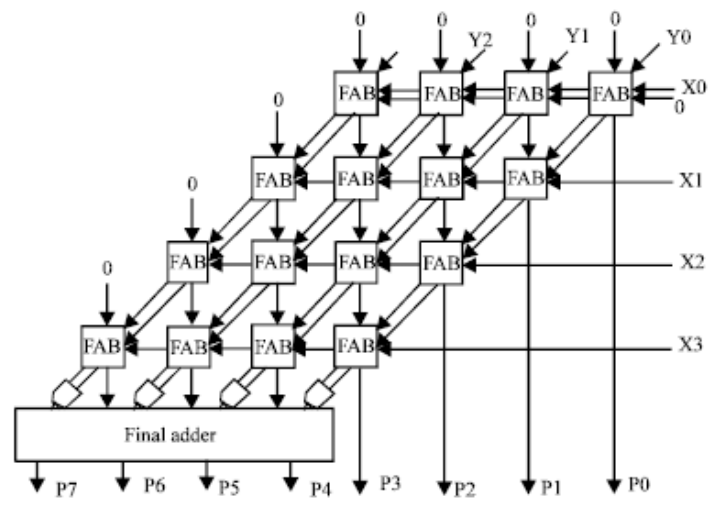


$$P_{x^\nu} = 10 \log_{10} \left(\frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 \right)$$

$$|_{dB} = P_{x^\nu} + 6.02\nu + 4.77$$

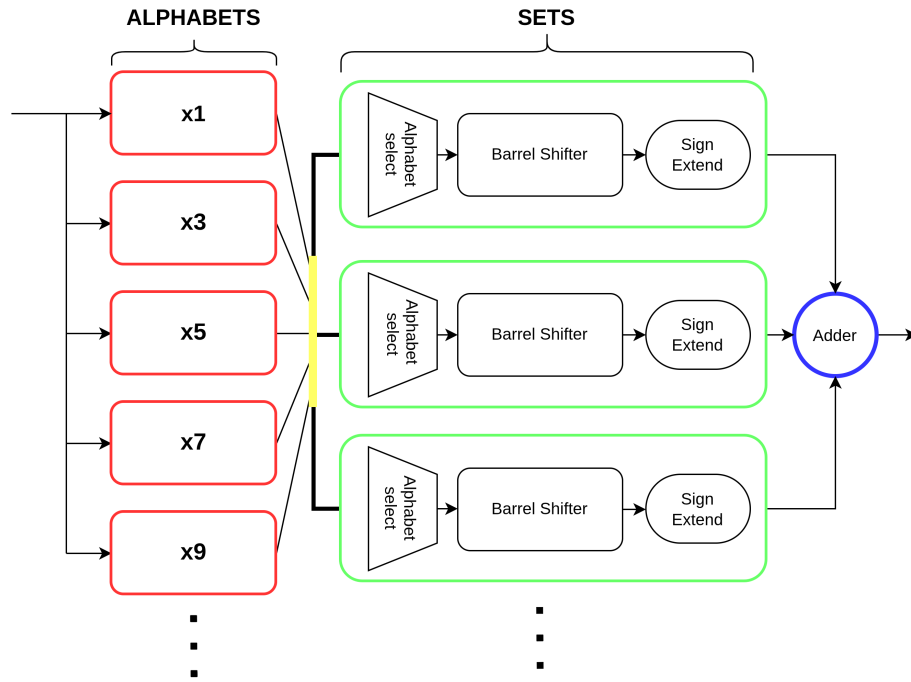
$$10 \log_{10}(1) = 0$$

ν	



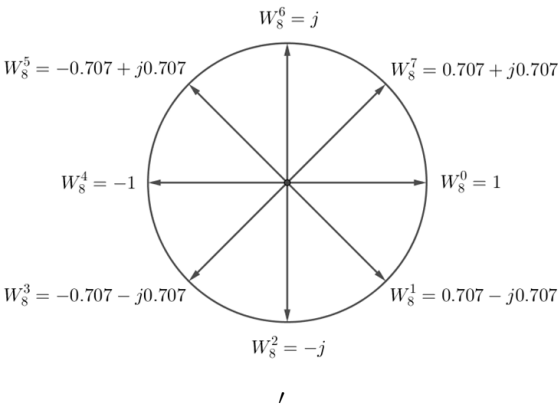
$\{1, 3, 5, 7\}$

$3x2x + x$



$$\{1, 3\}x27011011_201_210_211_2(1x \cdot 2^4) + (1x \cdot 2^3) + (3x \cdot 2^0)1x3x$$

$$\{1, 3, 5, 7\}x45101101_2101_2101_2(5x \cdot 2^3) + (5x \cdot 2^0)5x$$



Index	W_Real	W_Imag	Complex number
0	32768	0	1 + 0j
1	23170	-23170	0.7071 - 0.7071j
2	0	-32768	0 - 1j
3	-23170	-23170	-0.7071 - 0.7071j
4	-32768	0	-1 + 0j
5	-23170	23170	-0.7071 + 0.7071j
6	0	32768	0 + 1j
7	23170	23170	0.7071 + 0.7071j

$$W_N = e^{-j\frac{2\pi}{N}} N N$$

$$N$$

$$Q1.xx(-1,1)$$

$$BWA W = 1 + j0$$

$$ABW start$$

$$Out_1 Out_2 done$$

$$BW$$

$$stage = 1$$

$$B_{mult} \leftarrow B \ll (Tw Width - 1) \qquad \triangleright W = 1$$

$$SimpleMult = 1$$

$$B_{mult} \leftarrow (B, W) \qquad \triangleright$$

$$B_{mult} \leftarrow (B, W) \qquad \triangleright$$

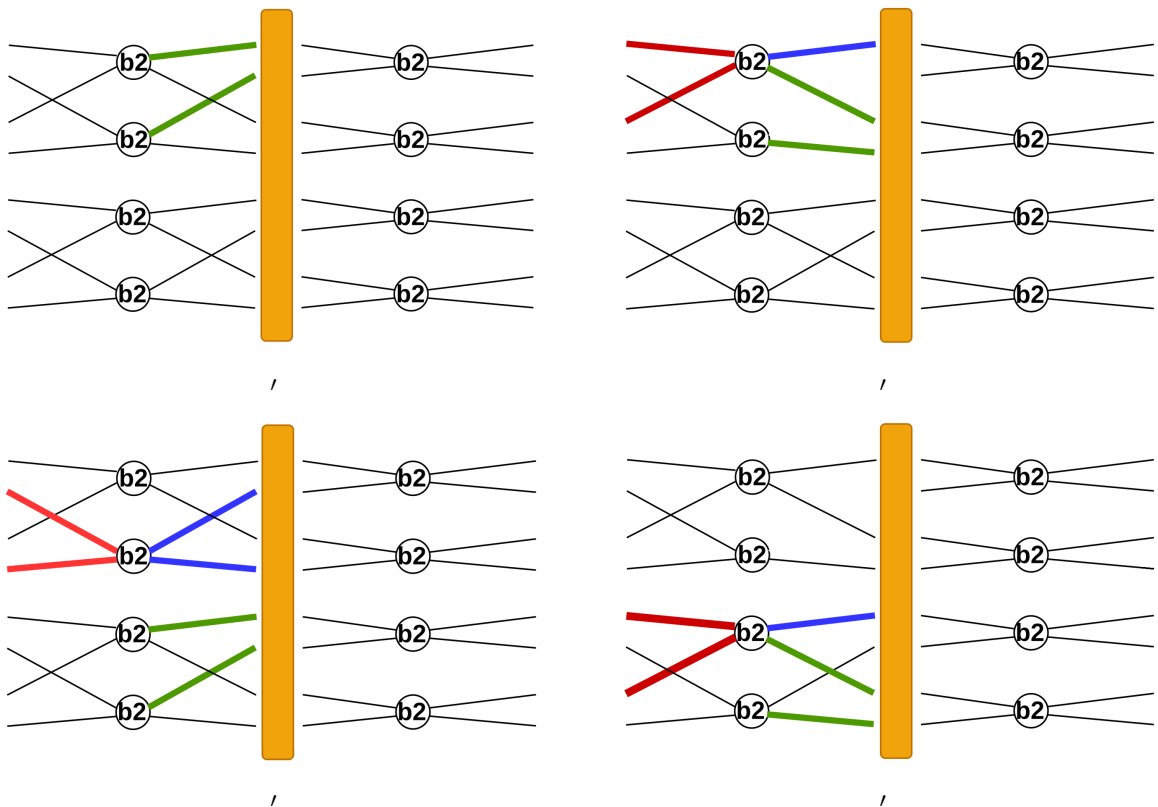
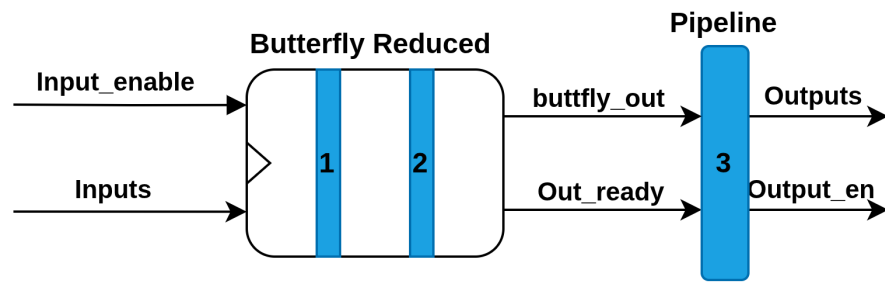
$$A_{delayed} \leftarrow (A, L_{mult}) \qquad \triangleright$$

$$done \leftarrow (start, L_{total}) \qquad \triangleright$$

$$B_{scaled} \leftarrow (B_{mult}, Tw Width - 1) \qquad \triangleright$$

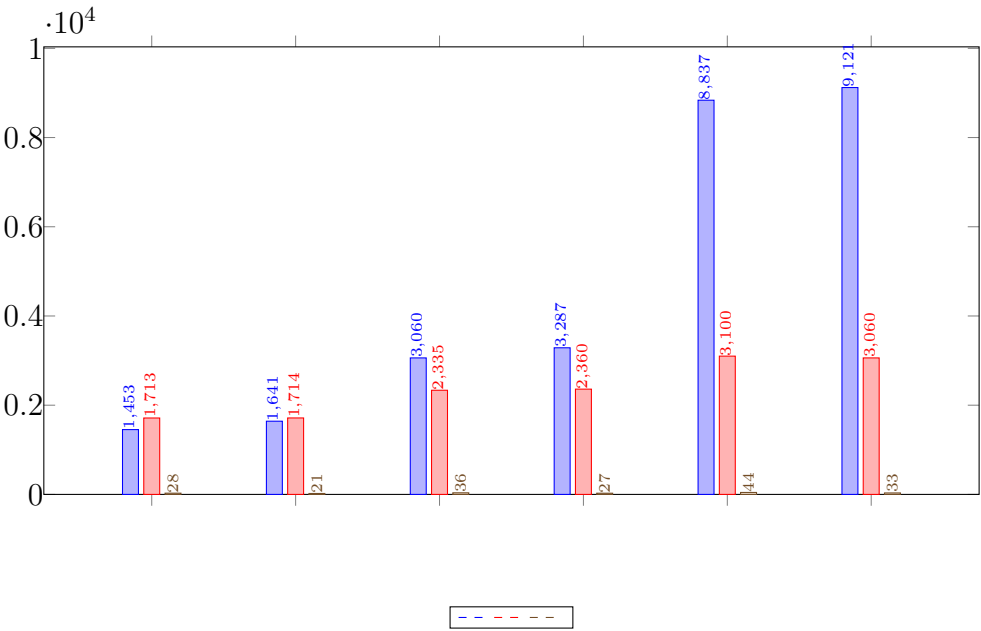
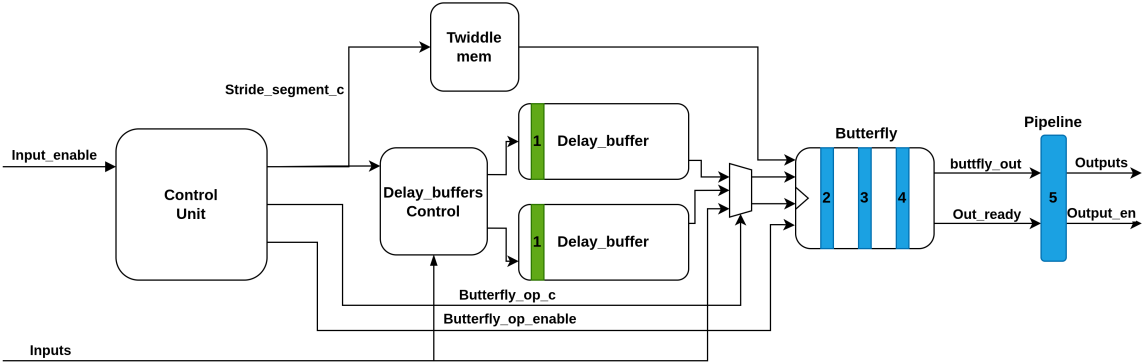
$$Out_1 \leftarrow A_{delayed} + B_{scaled}$$

$$Out_2 \leftarrow A_{delayed} - B_{scaled}$$

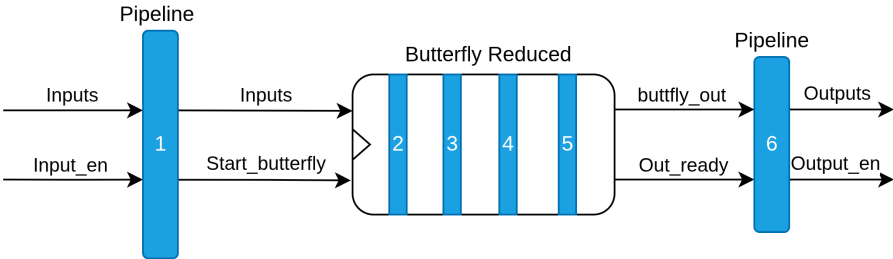


$$\text{DepthA} = 2^{\text{STAGENUM}-1}$$

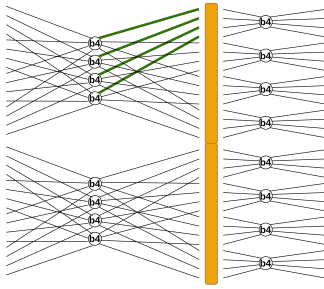
$$\text{DepthB} = \text{DepthA} \times 2$$



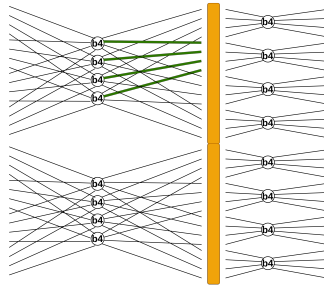
A



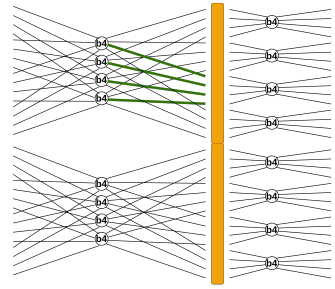
 $ABCDW_0W_1W_2start$ $Out_1Out_2Out_3Out_4done$ $BCDW_0W_1W_2$ $stagenum = 1$ $B_{mult} \leftarrow B \ll (Tw\ Width - 1)$ $\triangleright$ $C_{mult} \leftarrow C \ll (Tw\ Width - 1) D_{mult} \leftarrow D \ll (Tw\ Width - 1)$ $SimpleMult = 1$ $B_{mult} \leftarrow (B, W_0)$ $\triangleright$ $C_{mult} \leftarrow (C, W_1) D_{mult} \leftarrow (D, W_2)$ $B_{mult} \leftarrow (B, W_0)$ $\triangleright$ $C_{mult} \leftarrow (C, W_1) D_{mult} \leftarrow (D, W_2)$ $A_{delayed} \leftarrow (A, delaymult + 1)$ $\triangleright$ $done \leftarrow (start, delaymult + 3)$ $\triangleright$ $B_{scaled} \leftarrow (B_{mult}, Tw\ Width - 1)$ $\triangleright$ $C_{scaled} \leftarrow (C_{mult}, Tw\ Width - 1)$ $D_{scaled} \leftarrow (D_{mult}, Tw\ Width - 1)$ $t_0 \leftarrow A_{delayed} + C_{scaled}$ $t_1 \leftarrow A_{delayed} - C_{scaled}$ $t_2 \leftarrow B_{scaled} + D_{scaled}$ $t_3 \leftarrow B_{scaled} - D_{scaled}$ $Out_1 \leftarrow t_0 + t_2$ $Out_2 \leftarrow t_1 - j \cdot t_3$ $\triangleright t_{1r} + t_{3i}t_{1i} - t_{3r}$ $Out_3 \leftarrow t_0 - t_2$ $Out_4 \leftarrow t_1 + j \cdot t_3$ $\triangleright t_{1r} - t_{3i}t_{1i} + t_{3r}$



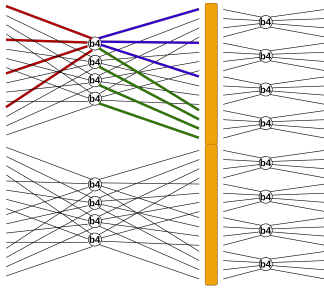
/



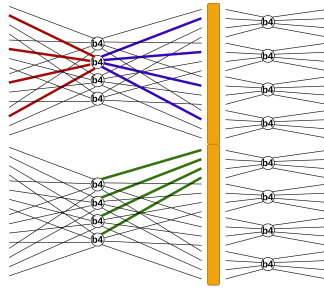
/



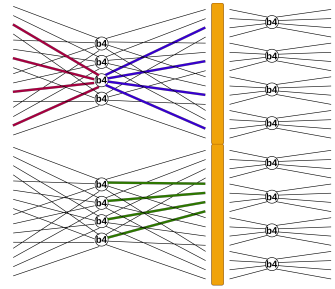
/



/



/



/